

AZ-104 – Lab 10: Implement Data Protection

Lab Objective (Big Picture)

This lab teaches **how Azure protects virtual machines from data loss and regional disasters** using:

- **Azure Backup** → *Data protection*
- **Azure Site Recovery (ASR)** → *Disaster recovery*

You learned:

- How backups are stored
 - How retention & soft delete work
 - Why deletion is intentionally hard
 - How replication across regions works
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Core Concepts You Must Understand

Concept	Purpose
Recovery Services Vault	Central store for backups & replication
Azure Backup	Protects VM data
Backup Policy	Controls frequency & retention
Soft Delete	Prevents accidental/malicious deletion
Restore Points	Snapshot recovery data
Azure Site Recovery	Cross-region DR solution

Architecture (Conceptual)

East US (Primary)

- └─ VM (az104-10-vm0)
 - └─ Recovery Services Vault (RSV-1)
 - └─ Backups

West US (Secondary)

└ Recovery Services Vault (RSV-2)

└ Replicated VM (ASR)

TASK-BY-TASK EXPLANATION

◆ Task 1: Deploy Infrastructure Using ARM Template

What You Did

- Deployed:
 - 1 Virtual Machine
 - 1 Virtual Network
- Using an ARM template

Why ARM Templates?

- ✓ Consistent deployments
- ✓ Infrastructure as Code (IaC)
- ✓ Exam favorite topic

Important Learning

- Region was controlled by:

"location": "[resourceGroup().location]"

→ Region comes from **resource group**, not VM directly.

◆ Task 2: Create Recovery Services Vault (RSV)

What is an RSV?

A **secure container** for:

- VM backups
- Site Recovery replication
- Policies

- Restore points

Key Settings

Setting	Meaning
Geo-redundant storage	Backup survives region failure
Soft Delete	Prevents accidental deletion
Cross-region restore	Restore from paired region

Important Exam Point

! RSV **must be in the same region as the VM** for Azure Backup.

◆ Task 3: Configure Azure VM Backup

Backup Workflow

VM → Snapshot → RSV → Retention

Backup Policy Explained

Setting	Why
Daily backup	Regular recovery points
Retain snapshots (2 days)	Fast restores
Vault retention	Long-term protection

What “Backup Pending” Means

- ✓ Backup scheduled
 - ✓ First backup not completed yet
 - ✗ NOT an error
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◆ Task 4: Monitor Azure Backup

Why Monitoring Matters

Backups can fail due to:

- VM shutdown
- Disk issues

- Network issues
- Quota issues

What You Configured

- Diagnostic logs → Storage Account
- Backup job tracking
- Site Recovery job logs

Key Exam Tip

Azure Backup integrates with:

- Log Analytics
- Storage
- Azure Monitor

◆ Task 5: Enable Azure Site Recovery (ASR)

What ASR Is (Very Important)

Azure Site Recovery ≠ Backup

Backup	Site Recovery
Point-in-time restore	Live VM replication
Data protection	Business continuity
Restore after failure	Failover during outage

ASR Workflow

Primary VM (Canada Central)

↓ Replication

Secondary Region (Canada East)

↓ Standby VM

Failover → VM becomes active

Why Two Vaults?

- RSV-1 → Backup
- RSV-2 → Disaster Recovery

Automation Account

Required for:

- Runbooks
- Replication orchestration
- Failover actions

Deletion Issues (VERY IMPORTANT REAL-WORLD)

Why You Couldn't Delete:

- RSV
- Restore point collections
- Resource group

Root Cause

Soft Delete (14 days)

Azure **locks backup data** to prevent:

- Ransomware
- Malicious admins
- Accidental deletion

Key Truth

Once a VM is backed up, deletion is delayed by design

Deletion Rules Summary

Action	Allowed?
Delete VM	✓
Stop backup	✓
Delete backup data	✓

Action	Allowed?
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Delete RSV immediately	✗
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Delete after retention	✓
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INTERVIEW QUESTIONS & ANSWERS

Q1: What is a Recovery Services Vault?

A: A storage entity used to manage backups and disaster recovery for Azure workloads.

Q2: Can we delete a Recovery Services Vault immediately?

A: No. If it contains backups, soft delete retention must expire first.

Q3: Difference between Azure Backup and Azure Site Recovery?

A: Backup protects data; Site Recovery provides business continuity through replication.

Q4: Why is Soft Delete mandatory?

A: To protect against accidental or malicious deletion, including ransomware.

Q5: What happens when you stop backup?

A: Backup stops, but data is retained until soft delete period ends.

Q6: Can restore points be manually deleted?

A: No. Azure manages restore point lifecycle.

Q7: Why is automation account required in ASR?

A: For replication orchestration, failover, and recovery workflows.